

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1 MH1100 Calculus I

Homework 2

Due before 6:00 pm, 09 November, 2023

(2 POINTS) Problem 1 Calculate the limit

(a)

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + ax} - \sqrt{x^2 + bx});$$

(b)

$$\lim_{x \rightarrow \infty} \frac{x^2 + 6x + 5}{2x^3 + x - 7}.$$

[Solution:] (a) We have

$$\begin{aligned} \lim_{x \rightarrow \infty} (\sqrt{x^2 + ax} - \sqrt{x^2 + bx}) &= \lim_{x \rightarrow \infty} (\sqrt{x^2 + ax} - \sqrt{x^2 + bx}) \cdot \frac{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \\ &= \lim_{x \rightarrow \infty} \frac{(x^2 + ax) - (x^2 + bx)}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \\ &= \lim_{x \rightarrow \infty} \frac{(a - b)x}{\sqrt{x^2 + ax} + \sqrt{x^2 + bx}} \\ &= \frac{a - b}{2} \end{aligned}$$

(b) We have

$$\lim_{x \rightarrow \infty} \frac{x^2 + 6x + 5}{2x^3 + x - 7} = 0$$

(2 POINTS) Problem 2 Find the limit

$$\lim_{x \rightarrow 0} \frac{2x^6 + 3x^7}{\sin(x) \cdot \sin(x^2) \cdot \sin(x^3)}.$$

[Solution:] Note that

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2x^6 + 3x^7}{\sin(x) \cdot \sin(x^2) \cdot \sin(x^3)} &= \lim_{x \rightarrow 0} \frac{x \cdot x^2 \cdot x^3}{\sin(x) \cdot \sin(x^2) \cdot \sin(x^3)} \cdot \frac{2x^6 + 3x^7}{x^6} \\ &= \lim_{x \rightarrow 0} \frac{x}{\sin(x)} \cdot \lim_{x \rightarrow 0} \frac{x^2}{\sin(x^2)} \cdot \lim_{x \rightarrow 0} \frac{x^3}{\sin(x^3)} \cdot \lim_{x \rightarrow 0} \frac{2x^6 + 3x^7}{x^6} \\ &= 2 \end{aligned}$$

Therefore we have

$$\lim_{x \rightarrow 0} \frac{2x^6 + 3x^7}{\sin(x) \cdot \sin(x^2) \cdot \sin(x^3)} = 2$$

(2 POINTS) Problem 3 Use a linear approximation to estimate the given number

$$\sin(92^\circ).$$

[Solution:] We assume $f(x) = \sin x$. At $x = a = 90^\circ = \frac{\pi}{2}$, we have $f(\frac{\pi}{2}) = 1$. The derivative is

$$f' = \cos x.$$

The linear approximation the function $f(x)$ at $x = \frac{92\pi}{180}$ is

$$L(x) = 1 - 0 \cdot \left(\frac{92\pi}{180} - \frac{\pi}{2}\right) = 1$$

thus we have

$$\sin(92^\circ) \approx 1.0$$

(1 POINTS) Problem 4 Find second derivative y'' by implicit differentiation for function $\sin y + xy = x^2$.

[Solution:] Note that

$$\begin{aligned} \cos y \cdot y' + y + xy' &= 2x \\ y' &= \frac{2x - y}{\cos y + x} \end{aligned}$$

Therefore we have

$$\begin{aligned} y'' &= \frac{(2 - y')(\cos y + x) - (2x - y)(-\sin y y' + 1)}{(\cos y + x)^2} \\ y'' &= \frac{2 \cos y + y - (2x - y)(-\sin y \frac{2x - y}{\cos y + x} + 1)}{(\cos y + x)^2} \\ y'' &= \frac{(2 \cos y - 2x + 2y)(\cos y + x) + (2x - y)^2 \sin y}{(\cos y + x)^3} \end{aligned}$$

(1 POINTS) Problem 5 If $f(1) = 10$ and $f'(x) \leq 1$ for $1 \leq x \leq 2$, how large can $f(2)$ possibly be?

[Solution:] From the Mean value theorem,

$$f(2) - f(1) = f'(c)(2 - 1)$$

since $f'(x) \leq 1$ for $1 \leq x \leq 2$, then $f'(c) \leq 1$, and

$$f(2) = f(1) + f'(c) \leq 10 + 1 = 11$$

Therefore $f(2)$ can be large as 11.

(1 POINTS) Problem 6 If $dx/dt = 5$ and $dy/dt = 4$, and

$$x^3 + y^3 + z^3 = 3$$

find dz/dt when $(x, y, z) = (1, 1, 1)$.

[Solution:] Take the derivative with respect to t at both sides, we have

$$x^2 dx/dt + y^2 dy/dt + z^2 dz/dt = 0$$

So we have

$$dz/dt = -\frac{x^2 dx/dt + y^2 dy/dt}{z^2}$$

Therefore, when $(x, y, z) = (1, 1, 1)$, we have

$$dz/dt = -\frac{1^2 * 5 + 1^2 * 4}{1^2} = -9$$

(1 POINT) Problem 7 Evaluate the limit

$$\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - 0.5}{\theta - \pi/3}.$$

(Hint: Definition of derivative)

[Solution:] We let $f(x) = \cos \theta$, and

$$\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - \cos(\pi/3)}{\theta - \pi/3} = f'(\theta)|_{\theta=\pi/3}$$

As we have

$$f'(\theta)|_{\theta=\pi/3} = -\sin \theta|_{\theta=\pi/3} = -\sqrt{3}/2$$

Therefore, we have

$$\lim_{\theta \rightarrow \pi/3} \frac{\cos \theta - 0.5}{\theta - \pi/3} = -\sqrt{3}/2$$